P180 | Axial-piston pump

Compact, water-lubricated axial-piston technology for a defined operating duty.

P180 supplies pressurised water to hydraulic actuators or a process. Select delivered flow at the working pressure, then match displacement, speed, drive power and inlet supply.

Published parameter	P180 range rating
Maximum displacement (cm³/rev)	225.0
Minimum displacement (cm³/rev)	135.0
Maximum speed, unboosted (rpm)	Not listed
Maximum speed, boosted (rpm)	2000
Maximum input power (kW)	114
Maximum water flow (L/min)	430
Maximum continuous pressure (bar)	160
Mass (kg)	82
Maximum temperature (°C)	50
Minimum temperature (°C)	2

Nominal ordering displacements: 134 / 180 / 225 cm³/rev. Nominal options and range-table limits are different source entries; use the selected configuration.

Design & environmental advantages

Multiple pistons smooth the delivery of power. Water lubricates the working surfaces, removing hydraulic oil from this circuit. Standard stainless construction and selectable internal materials support fluid-specific selection.

Efficiency & selection

Match component size to useful work. Read the supplied efficiency curve at pressure and speed. Avoid sustained relief flow and excessive throttling; these convert useful hydraulic power into heat.

Installation and procurement

Specify flow, pressure, speed, fluid chemistry, inlet supply, rotation, seal type, shaft coupling and mounting. No induced radial or axial shaft load. Nominal 10 µm / 25 µm absolute filtration, beta10 ≥ 75, is listed in the pump catalogue. Variable versions are listed for P60 and P180 only.

The range table lists a minimum of 135 cm³/rev; nominal ordering and some curve labels use 134. Match the quotation to its approved displacement and drawing.



Family photograph; configuration varies.

Connect the selected design

Two-port pumps need flooded, positive inlet pressure and cannot lift water. Three-port boosted units separate the feed and case drain. Specify that arrangement in the order.

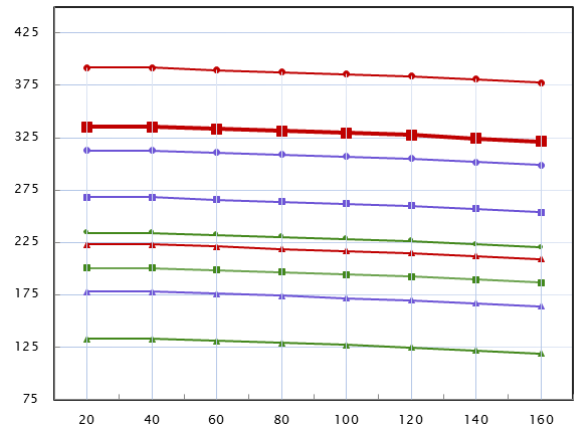
Reference conversions: 160 bar = 2,321 psi; 1 L/min = 0.264172 US gpm; 1 kg = 2.20462 lb.

Janus Axial Piston Water Pump - V.01.25 (3).pdf, pp. 2-4. Website product context checked 17 Oct 2026.

P180 | Performance curves

Read displacement, speed and pressure together. Published curve data are preserved.

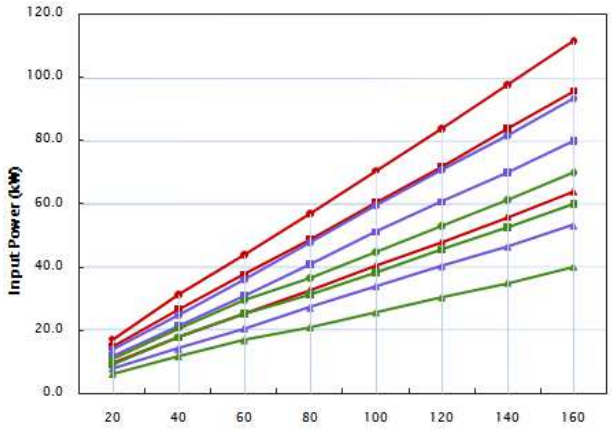
Output flow (L/min)



Pressure (bar)

225cc 1750rpm 225cc 1500rpm 225cc 1000rpm
180cc 1750rpm 180cc 1500rpm 180cc 1000rpm
135cc 1750rpm 135cc 1500rpm 135cc 1000rpm

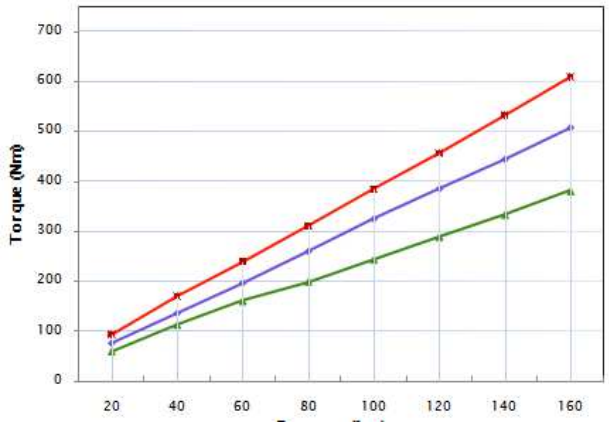
Input power (kW)



Pressure (bar)

225cc 1750rpm 225cc 1500rpm 225cc 1000rpm
180cc 1750rpm 180cc 1500rpm 180cc 1000rpm
135cc 1750rpm 135cc 1500rpm 135cc 1000rpm

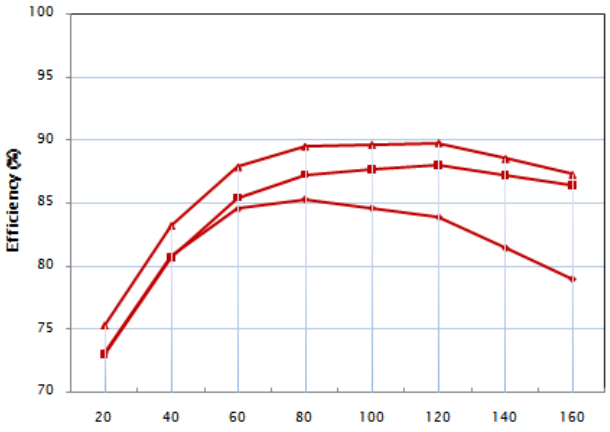
Input torque (N·m)



Pressure (bar)

225cc 1500rpm 180cc 1500rpm 135cc 1500rpm

Overall efficiency (%)



Pressure (bar)

225cc 500 rpm 225cc 1000rpm 225cc 1500rpm

Apply the curves to the duty

The range table lists a minimum of 135 cm³/rev; nominal ordering and some curve labels use 134. Match the quotation to its approved displacement and drawing.

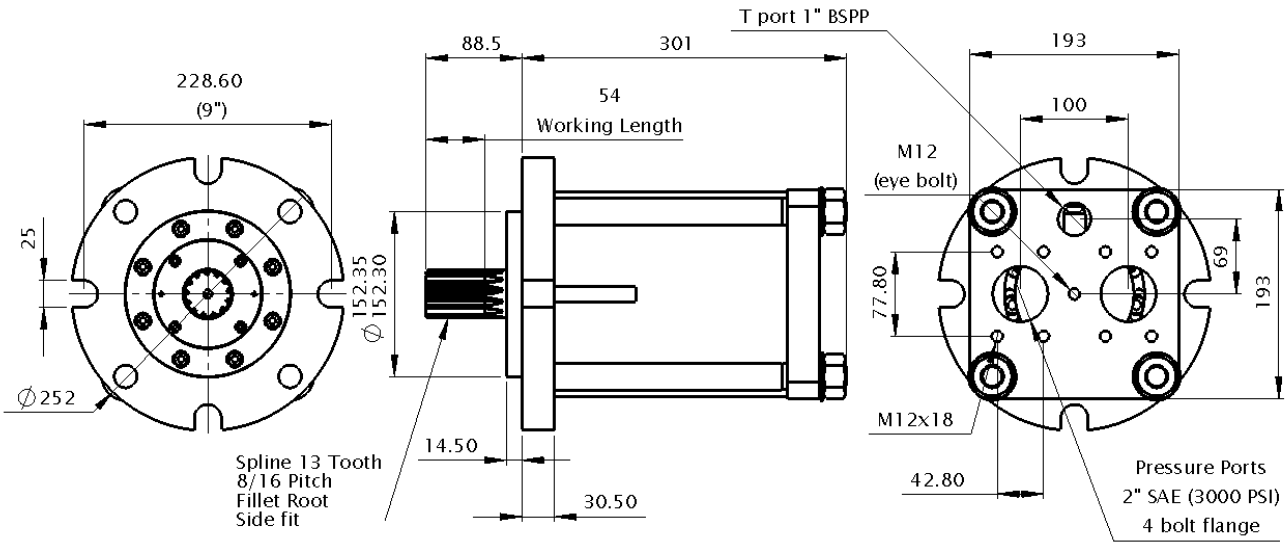
1 L/min = 0.264172 US gpm; 1 bar = 14.5038 psi; 1 kW = 1.34102 hp; 1 N·m = 0.737562 lbf·ft.

Janus Axial Piston Water Pump - V.01.25 (3).pdf, p. 18.

P180 | Dimensions & connections

Manufacturer technical reference | dimensions govern; reproduction is not to scale.

PB160-135W, PB160-180W & PB160-225W



Note:- The P180 pump is only supplied in the 3 port Boosted configuration with mechanical seal.

Port threads and flange classes are connection designations, not converted diameters or operating-pressure ratings. Select the exact configuration shown on the approved order drawing.